

Timeformer: A Controlled Diagnostic Study of Temporal Semantic Traceability

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Abstract. Words change meaning over time, yet most computational models treat token representations as period-independent. We study *temporal traceability*: whether the same surface token, queried at different periods, occupies period-appropriate neighborhoods. A model may improve predictive accuracy with temporal information while still assigning nearly identical neighborhoods across periods; this gap is our focus.

We introduce Timeformer as a family of token-time conditioning designs. Standard (no time) and Additive (global time vector) are external baselines. The Token-Time Transformer exposes token identity and period jointly before self-attention; a Memory-Augmented extension adds causal prototype memory. All models are trained with masked language modeling on a synthetic benchmark with known ground-truth trajectories.

The main finding is that temporal conditioning of any form produces substantially larger neighborhood drift than no conditioning: conditioned models reach $\Delta \approx -0.56$ for drifting subjects versus -0.22 for Standard, stable across 31 seeds. Additive and Token-Time are statistically indistinguishable on drift score; Token-Time shows a reproducible advantage only on the more sensitive ambiguous probe diagnostic (0.595 vs. 0.579). The Memory-Augmented extension shows no robust improvement; oracle analysis identifies mean-prototype aggregation as the failure point.

Keywords: Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; today it describes a social media post reaching millions of users. Most language models, however, take no account of when a text was written: a word’s representation depends on its surrounding context, but not on the historical period in which it appeared.

The ability to query a word at a specific period matters beyond technical benchmarks. Whether the shift of *gay* from *cheerful* to its contemporary sense involved movement toward or away from stigmatized terms [3] is an empirical question that requires `gay@1920` and `gay@1980` as directly comparable objects in a shared space.

The dominant computational approach, lexical semantic change detection (LSCD), measures such shifts by training distributional representations on time-sliced corpora and comparing them across periods [6, 7]. One model may associate *broadcast* in 1900 with *sow* and *field*, while another associates it with *signal* and *audience*; change is inferred from the distance between the two spaces. It summarizes the shift as a single similarity score between separately trained models, not as a queryable `token@time` object.

A subtler limitation arises when a single model receives temporal information during training. A model may correctly interpret *virus* as biological infection in 1900 and as malicious software in 2020, improving word-prediction accuracy, while still assigning nearly identical neighborhoods across periods. Prediction accuracy and geometric traceability are separable properties.

We call a representation *temporally traceable* if the same surface token, queried at different periods, occupies neighborhoods consistent with its meaning at that period, making `token@time` a directly inspectable object. Timeformer evaluates this property across the conditioning family introduced above in a controlled synthetic benchmark where each subject trajectory is known.

The paper makes four contributions: it formulates temporal traceability as a `token@time` representation problem distinct from period-label recoverability; shows that temporal conditioning of any form produces substantially larger neighborhood drift than no conditioning, while Additive and Token-Time are not robustly distinguishable on drift score; evaluates traceability with direct subject-level geometric diagnostics; and characterizes mean-pooled prototype memory as a diagnostic tool rather than a robust improvement.

2 Related Work

Lexical semantic change detection (LSCD) is the dominant computational framing for meaning shift. Diachronic word embeddings [3] learn one distributional space per time slice, align independently trained models, and measure movement in neighborhood structure. We share the intuition that semantic change is geometric, but not the architecture: LSCD yields a scalar comparison between spaces rather than a jointly parameterized `token@time` representation. Standard benchmarks formalize this task [6, 7], while also inheriting natural-corpus confounds such as frequency and genre drift. Our controlled benchmark plants trajectories so that traceability can be attributed to model design rather than corpus composition.

Temporal neural language models avoid separate spaces by conditioning one model on time. Prior work uses timestamps for knowledge-intensive QA, masked language modeling, or period-stratified downstream evaluation [1, 5, 4]. These results show that temporal conditioning can improve period-appropriate prediction, but they do not test whether the same token occupies different neighborhoods at different periods in a shared space.

Linear probes provide another partial view. A period probe measures whether time is recoverable from a hidden state, not whether a token has moved to a dif-

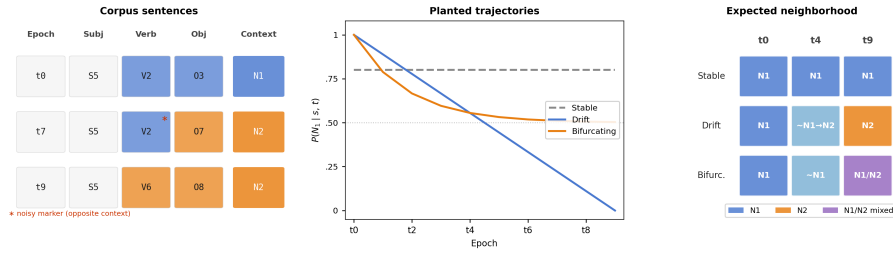


Fig. 1. Benchmark design overview. *Left:* SVO sentences for a drifting subject across three periods; an N_1 verb appearing in an N_2 sentence at $t7$ (marked *) illustrates the deliberate noise in local markers. *Center:* planted $P(N_1 | s, t)$ trajectories for Stable (flat), Drift (monotonic), and Bifurcating (levelling at 0.5) subjects across $t0$ – $t9$. *Right:* expected nearest-neighbor behavior for a traceable representation.

ferent semantic neighborhood. We therefore report probe accuracy as auxiliary evidence and treat neighborhood movement as the primary criterion for traceability.

Finally, semantic change may involve bifurcation rather than monotonic drift. Giulianelli et al. show that words can develop coexisting senses [2]; mean-pooled representations then collapse both senses into a centroid that represents neither. Our stable, drifting, and bifurcating trajectories test this distinction directly, and the memory extension exposes where mean-pooled prototypes fail.

3 Controlled Temporal Benchmark

The benchmark gives each subject a known temporal trajectory, making it possible to ask directly whether a model’s representation of a subject at $t0$ has moved toward the expected neighborhood by $t9$. This distinguishes it from natural-language LSCD benchmarks, where ground truth is often annotated or inferred post-hoc and corpus confounds are hard to separate from genuine semantic change.

The corpus uses a subject–verb–object (SVO) grammar: every sentence consists of one subject, one verb, and one object. Two semantic neighborhoods, N_1 and N_2 , partition the vocabulary of 8 verbs and 8 objects: N_1 uses $\{V1, \dots, V4\}$ and $\{O1, \dots, O4\}$; N_2 uses $\{V5, \dots, V8\}$ and $\{O5, \dots, O8\}$. A sentence generated from N_1 therefore looks like S5 V2 O3, while one from N_2 looks like S5 V6 O8; the subject token is the same in both cases, but its co-occurring words differ. Intuitively, the intended meaning of subject s at period t is how often it appears in N_1 sentences; formally, $P(N_1 | s, t)$ is the probability that a sentence is drawn from N_1 . The vocabulary contains 30 subjects; each subject’s trajectory is determined by how $P(N_1 | s, t)$ evolves across the 10 periods $t0$ through $t9$.

Figure 1 (center) shows the three planted trajectory classes. Stable subjects maintain an approximately constant $P(N_1 | s, t)$ throughout. Drift subjects follow a monotonic trajectory from near-exclusive N_1 ($P(N_1 | s, t) \approx 1$ at $t0$) to

near-exclusive N_2 ($P(N_1 | s, t) \approx 0$ at t_9): a drifting subject that appears in sentences like **S5 V2 O3** early on will appear in sentences like **S5 V6 O8** by the final period. Bifurcating subjects begin in N_1 and move toward a mixed state ($P(N_1 | s, t) \approx 0.5$) where both neighborhoods appear with equal frequency, approximating two coexisting senses. The benchmark contains 10 subjects per class.

Each local marker is drawn from the opposite neighborhood 25% of the time (Figure 1, left), so the model cannot recover the planted neighborhood from co-occurring words alone — it must use subject identity together with the period. The `true_context` label is withheld from training and used only for evaluation.

The choice of a synthetic benchmark is deliberate. Natural-language corpora introduce confounders — frequency change, genre shift, and subcorpus composition — that make it impossible to attribute a representational effect to model design rather than data artefacts. By planting trajectories with known $P(N_1 | s, t)$, we can ask directly whether a model’s geometry matches the generating process. The cost is that results are mechanistic diagnostics, not claims about open-domain diachronic language; this scope is revisited in the Discussion.

The benchmark supports several evaluation conditions. The standard split uses the same 75% marker accuracy as training. Two harder splits force the verb, or both verb and object, to come from the opposite context. An ambiguous split sets fidelity to 50%, so local markers carry no reliable signal; only subject identity and period remain. Two additional conditions test temporal behaviour directly and are described in Section 5: the contrastive set presents the same sentence under two different period labels to check whether changing only the period changes the preferred context; the continuation set uses periods held out from training to test generalisation.

4 Timeformer Architecture

Timeformer is designed to make a subject representation depend explicitly on the queried period. We study this through a controlled comparison: Standard has no period input, Additive injects a global period vector, Token-Time exposes token identity and period jointly before self-attention, and the Memory-Augmented extension adds causal historical prototypes before the encoder.

All models follow the same outer pipeline. A sentence is tokenized as **[CLS] S V O [SEP]**. Each token w_i is mapped to an input embedding $z_i \in \mathbb{R}^d$. The sequence $(z_1, \dots, z_n)$ is processed by a Transformer encoder, and the final hidden state at the subject position,

$$(h_1, \dots, h_n) = \text{Encoder}(z_1, \dots, z_n), \quad h_s = h_{\text{subj}},$$

is the `token@time` representation inspected by the diagnostics. All models are trained with masked language modeling (MLM): some tokens are masked, and the model must reconstruct them from the remaining tokens and, when available, the period input. What changes across the design comparison is how the subject’s input embedding z_s is built before the encoder sees the sentence.

Figure 2. Controlled design comparison



Fig. 2. Controlled design comparison. **Standard:** Transformer baseline with no temporal signal. **Additive:** temporal baseline that adds the same period vector to every token. **Token-Time:** Token-Time Transformer using joint token-time projection. **Memory-Augmented:** extension of Token-Time with Prototype Memory and Temporal Attention. New components are highlighted in yellow; dashed grey boxes mark inactive components.

The temporally informed models use a shared period vector $\tau(t) \in \mathbb{R}^d$. It is a learned projection of sinusoidal features of t/T , where T is the total number of training periods. In plain terms, $\tau(t)$ is the vector representation of the current period. It is distinct from the position encoding p_i , which only tells the Transformer whether a token is in subject, verb, or object position. Additive and Token-Time differ in how they combine $\tau(t)$ with token identity; the Memory-Augmented extension then adds causal memory on top of Token-Time. Figure 2 summarizes the comparison.

4.1 Baseline: Standard Transformer

Standard is the null hypothesis: no period signal is available, so any temporal behavior must arise from changes in the observed words alone.

$$z_i = e(w_i) + p_i.$$

4.2 Baseline: Additive Time-Conditioned Transformer

The Additive baseline asks whether knowing the current period is enough. It adds the same period vector $\tau(t)$ to every token:

$$z_i = e(w_i) + p_i + \tau(t).$$

Now **S5 V2 O3** at t_0 and t_9 are distinguishable. The subject embedding is $e(\mathbf{S5}) + p_1 + \tau(0)$ at t_0 , and $e(\mathbf{S5}) + p_1 + \tau(9)$ at t_9 . However, the same temporal change is also applied to the verb and object: **V2** receives $e(\mathbf{V2}) + p_2 + \tau(t)$, and **O3** receives $e(\mathbf{O3}) + p_3 + \tau(t)$. Thus Additive tells the whole sentence which period it belongs to, but it does not explicitly bind period to each token’s identity. If Additive succeeds, global period information is sufficient. If it does not, the next question is whether period information must be exposed jointly with each token before contextualization.

4.3 Token-Time Transformer

The Token-Time Transformer in Figure 2 asks whether exposing token identity and period jointly before self-attention is enough. Instead of adding the same period vector to all tokens, Token-Time concatenates each token embedding with $\tau(t)$ and projects the pair:

$$z_i = W[e(w_i); \tau(t)] + p_i,$$

where $e(w_i), \tau(t) \in \mathbb{R}^d$ and $W \in \mathbb{R}^{d \times 2d}$. A linear W does not by itself create nonlinear interaction; rather, it gives the encoder token-period inputs before self-attention, normalization, and later nonlinear transformations. For traceability, **S5** at t_0 and **S5** at t_9 therefore enter the encoder as distinct period-conditioned inputs.

4.4 Memory-Augmented Extension

The Memory-Augmented extension asks whether causal history helps beyond token-time conditioning. It starts exactly like Token-Time: for a current sentence containing **S5** at t_7 , the model first builds token-time input embeddings for **[CLS S5 V O [SEP]]**. It then modifies only the subject embedding before the Transformer encoder sees the sentence.

The additional information comes from Prototype Memory. The extension stores one prototype vector per subject and past period. A prototype $m(s, t)$ is the average representation of subject s across training sentences from period t . The memory is causal: at t_7 , the model may consult $\{m(\mathbf{S5}, 0), \dots, m(\mathbf{S5}, 6)\}$, but not prototypes from t_7 or later.

Temporal attention turns this list of past prototypes into a history summary for the current subject. Let z_s be the current Token-Time subject embedding and let M_s be the matrix of available causal prototypes, one row per past period. The attention operation uses z_s as the query and M_s as keys and values:

$$r_s = \text{Attn}(z_s, M_s, M_s).$$

The vector r_s summarizes the parts of the subject’s history that are most relevant to the current period. A learned scalar gate g then decides how much this history summary should change the current subject embedding:

$$\tilde{z}_s = z_s + g \cdot (\text{LN}(z_s + r_s) - z_s).$$

Here, LN denotes layer normalization, and g is a learned scalar gate initialized at zero so that the extension reduces to Token-Time at initialization. Finally, $\tilde{z}_s$ replaces z_s in the input sequence; verb and object embeddings are unchanged, so the subject is the only token that enters the encoder informed by causal prototype history.

5 Evaluation Protocol and Diagnostics

All models in the comparison chain are trained with masked language modeling over the synthetic SVO corpus. The planted `true_context` label is withheld from training and used exclusively for post-hoc evaluation: it tests whether the learned subject representation reflects the planted temporal trajectory. The primary evaluation target is h_s , the final hidden state at the subject position, because subjects are the lexical items whose meanings are designed to drift, remain stable, or bifurcate over time.

We use four diagnostics, each targeting a distinct aspect of temporal representation.

D1 – Linear probe. Asks whether h_s encodes the planted context label at all. We train a linear classifier on h_s from the standard test split to predict `true_context` and evaluate it on each temporal split. High accuracy indicates that period-appropriate context is linearly separable in the representation, but does not reveal whether that separation shifts systematically over time.

D2 – Context drift score. Asks whether the neighborhood of a subject shifts in the expected direction across periods. For each period and subject class, we retrieve the $k = 10$ nearest neighbors of h_s by cosine similarity (excluding the query itself) and record the proportion assigned to N_1 . A representation is traceable at period t if this proportion tracks $P(N_1 \mid s, t)$ — the same probability that governs corpus generation at that period. For drifting subjects, the proportion should therefore decrease monotonically from t_0 to t_9 ; for stable subjects it should remain near its initial value; for bifurcating subjects it should decrease less steeply, as the representation splits between neighborhoods rather than migrating from one to the other.

D3 – Contrastive diagnostic. Asks whether the period label alone drives representational change. We present the same sentence with two different period labels and compute the sign-flip rate: the proportion of pairs for which swapping the period reverses the nearest-neighbor assignment, from N_1 -dominant to N_2 -dominant or vice versa. A model that ignores the period input produces a sign-flip rate of zero; a model that responds to it produces a rate above zero.

D4 – Continuation diagnostic. Asks whether the learned temporal mapping generalizes beyond training. We evaluate on held-out periods t_8 – t_9 , excluded from MLM training. A model that has internalized the direction of change, rather than memorizing period-specific patterns, should maintain coherent neighborhood assignments in these unseen periods.

The four diagnostics are interpreted through a fixed comparison chain. Additive vs. Standard isolates the effect of adding any temporal signal. Token-Time vs. Additive tests whether exposing token identity and period jointly before self-attention is more effective than applying the period globally as a uniform shift. Memory-Augmented vs. Token-Time tests the contribution of causal prototype memory, the mechanism most expected to affect D3 and D4.

For the Memory-Augmented extension, we include three controls. In the *shuffled-memory* control, each subject receives prototypes drawn from a different subject, disrupting identity-specific history while preserving the memory pathway. In the *no-history* control, the memory path is present but empty. In the *oracle-memory* diagnostic, learned prototypes are replaced by fixed vectors encoding the true historical $P(N_1 \mid s, t)$ trajectory. The extension passes the evaluation criterion if it outperforms both the Token-Time Transformer and the two controls; the oracle diagnostic then indicates how much headroom remains relative to a perfect memory signal.

6 Results

We report means and 95% confidence intervals over 31 training seeds. The main finding is that any temporal conditioning produces substantially larger neighborhood drift than no conditioning; Additive and Token-Time are not robustly distinguishable on drift score. The Memory-Augmented extension does not provide a robust improvement over the Token-Time Transformer.

6.1 Temporal Neighborhood Movement

Figure 3 and Table 1 show the context drift score across periods. Standard is a sanity check: its weak movement ($\Delta = -0.221$, 95% CI $[-0.240, -0.203]$) confirms that the benchmark alone does not produce the stronger trajectories seen when period information is available. All three conditioned models reach $\Delta \approx -0.56$, with overlapping confidence intervals; the gap between any conditioned model and Standard is the primary geometric evidence for the effect of temporal conditioning.

The class-specific curves serve as controls (Table 1). Stable subjects show much smaller movement and no coherent N_1 -to- N_2 trend. Bifurcating subjects show intermediate drift consistent with a trajectory toward a mixed state, not a complete transition. The geometric diagnostic distinguishes all three trajectory classes without relying on probe accuracy alone.

6.2 Probe and Contrastive Diagnostics

The linear probe results reveal a distinction that drift score does not. On the ambiguous split, where local verb and object markers are uninformative, Standard reaches 0.565 [0.561, 0.568], Additive 0.579 [0.575, 0.584], Token-Time 0.595

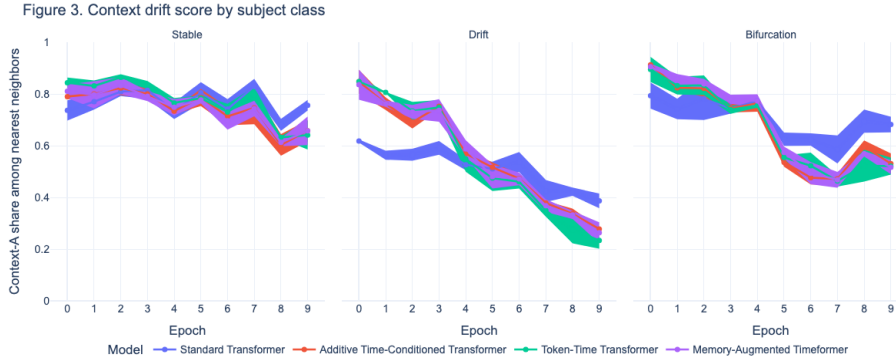


Fig. 3. Context drift score across periods. All conditioned models move drifting subjects from N_1 toward N_2 substantially more than Standard; Additive, Token-Time, and Memory-Augmented produce similar trajectories. Stable and bifurcating subjects provide class-specific controls.

[0.589, 0.601], and Memory-Augmented 0.598 [0.590, 0.606]. The confidence intervals for Additive and Token-Time do not overlap, showing a reproducible advantage for joint token-time conditioning on this more sensitive diagnostic even when drift scores are statistically indistinguishable. Probe accuracy alone is insufficient to establish temporal traceability — it measures recoverability of the planted label, not neighborhood geometry — but it captures a different aspect of how temporal information is encoded.

The contrastive sign-flip rates are similar across conditioned models: Standard 0.000, Additive 0.602 [0.549, 0.655], Token-Time 0.624 [0.576, 0.673], Memory-Augmented 0.587 [0.536, 0.638]. Confidence intervals overlap, confirming that all conditioned models respond to period information at roughly the same rate in contrastive pairs.

6.3 Memory Diagnostics

The Memory-Augmented extension does not produce a robust gain over the Token-Time Transformer. Across 31 seeds, learned prototypes reach a continuation accuracy of 0.755 [0.750, 0.760], marginally below Token-Time at 0.763 [0.759, 0.766] ($\Delta = -0.008$). The oracle diagnostic shows the limits of this architecture: replacing learned prototypes with fixed vectors encoding the true $P(N_1 | s, t)$ trajectory yields a continuation of 0.769 [0.761, 0.777] and a sign-flip rate of 0.694 [0.576, 0.812], versus 0.624 for Token-Time. The sign-flip improvement ($\Delta = +0.070$) is larger than the continuation improvement ($\Delta = +0.007$), but both CIs are wide. The dissociation has a direct explanation: the MLM objective optimizes token prediction, not trajectory geometry; perfect historical prototypes can therefore improve contrastive behavior — which depends on lo-

Table 1. Context drift score at the first and last periods. Values are means over 31 seeds; $\Delta = t9 - t0$; 95% CI half-widths in brackets.

Class	Model	t0	t9	Δ [CI $_{\pm}$]
Stable	Standard	0.741	0.781	+0.040 [± 0.017]
	Additive	0.827	0.678	-0.150 [± 0.027]
	Token-Time	0.835	0.655	-0.180 [± 0.024]
	Mem-Aug.	0.830	0.672	-0.158 [± 0.021]
Drift	Standard	0.632	0.410	-0.221 [± 0.018]
	Additive	0.839	0.272	-0.567 [± 0.019]
	Token-Time	0.833	0.269	-0.563 [± 0.020]
	Mem-Aug.	0.833	0.274	-0.559 [± 0.022]
Bifurc.	Standard	0.793	0.660	-0.133 [± 0.018]
	Additive	0.895	0.530	-0.365 [± 0.017]
	Token-Time	0.891	0.525	-0.366 [± 0.016]
	Mem-Aug.	0.878	0.530	-0.348 [± 0.015]

Table 2. Main traceability diagnostics (means, 31 seeds). Drift: context drift score Δ for drifting subjects; Flip: contrastive sign-flip rate (D3); Ambig./Cont.: linear-probe accuracy on h_s . 95% CI in brackets.

Model	Drift	Flip	Ambig.	Cont.
Standard	-0.221 [± 0.018]	0.000	0.565 [± 0.004]	0.730 [± 0.004]
Additive	-0.567 [± 0.019]	0.602 [± 0.053]	0.579 [± 0.005]	0.763 [± 0.003]
Token-Time	-0.563 [± 0.020]	0.624 [± 0.049]	0.595 [± 0.006]	0.763 [± 0.004]
Mem-Aug.	-0.559 [± 0.022]	0.587 [± 0.051]	0.598 [± 0.008]	0.755 [± 0.005]

cal period sensitivity — but cannot substitute for a training signal that directly rewards geometric continuity across unseen periods.

7 Discussion and Conclusion

The controlled benchmark gives a clear two-level answer. At the first level, temporal conditioning of any form produces substantially larger neighborhood drift than no conditioning: Standard reaches $\Delta = -0.221$ [$-0.240, -0.203$] for drifting subjects, while all conditioned models cluster near $\Delta \approx -0.563$ with overlapping 95% confidence intervals, stable across 31 seeds. This is the main geometric result.

At the second level, the form of conditioning matters less than its presence. Additive and Token-Time are statistically indistinguishable on drift score; Token-Time shows a reproducible advantage only on the ambiguous probe (0.595 vs. 0.579, non-overlapping CIs), suggesting that joint token-time conditioning encodes temporal information more precisely in the subject representation, even when the aggregate neighborhood trajectory is similar. In this benchmark, where semantic change is concentrated in a fixed subject position and the vocabulary

Table 3. Memory diagnostics (means, 31 seeds). Continuation (D4): linear-probe accuracy on held-out periods $t8-t9$; Flip: contrastive sign-flip rate (D3); 95% CI half-widths in brackets.

Model	Cont.	Flip	Δ cont. vs. Token-Time
Token-Time (ref.)	0.763 $[\pm 0.004]$	0.624 $[\pm 0.048]$	–
Mem-Aug. learned	0.755 $[\pm 0.005]$	0.587 $[\pm 0.051]$	–0.008
Mem-Aug. oracle	0.769 $[\pm 0.004]$	0.694 $[\pm 0.059]$	+0.007 $[\pm 0.007]$

is small, both additive and joint period injection are sufficient to produce the observed neighborhood movement.

The negative result for the Memory-Augmented extension is specific to mean-pooled prototypes in this controlled setting and should not be read as a general verdict on causal memory. The oracle diagnostic confirms this: perfect historical prototypes improve the sign-flip rate by +0.070 over Token-Time (0.694 vs. 0.624), showing the memory pathway is useful in principle even if the effect is variable across seeds. The failure of learned prototypes is a failure of aggregation: for bifurcating subjects, which develop two coexisting senses in later periods, the mean prototype is a centroid between two clusters, representing neither well.

Several limitations bound these conclusions. The probe-level advantage of Token-Time over Additive is small and may not generalize beyond controlled positions, small vocabulary, and semantic change concentrated in a known lexical class; it does not appear on the drift score at all. The MLM objective optimizes token prediction rather than trajectory geometry, so traceability emerges as a side effect rather than a direct training target. Finally, the diagnostic suite was co-designed with the architecture family, limiting its role as an independent evaluation for models outside this design space.

Designing memory mechanisms that preserve the internal structure of historical representations — rather than collapsing them to a single prototype — is the most direct next step. Evaluating token-time conditioning on natural corpora with attested semantic change is the necessary path toward claims beyond this controlled setting.

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